

New Cloud Database Tools Help Vodafone Fiji Make Better Decisions

Vodafone Fiji Limited is a 100 percent locally owned wireless telecommunications services provider, with more than 760,000 subscribers on its network, 260 employees, and revenue of U.S. \$185 million. Vodafone Fiji works closely with the Vodafone Group, a British multinational telecommunications conglomerate, which predominantly services Asia, Africa, Europe, and Oceania, and other operating companies to deliver cutting-edge technology to the people of Fiji. It has 85 percent of that country's market share in telecommunications services.

Prepaid customers account for 96 percent of Fiji's mobile communications market. (Less than 25 percent of U.S. mobile customers use prepaid services.) Prepaid customers are not bound by mobile service contracts, and thus are always looking for better ways to save. Consequently, Vodafone Fiji has to come up with better deals for its customers to persuade them to continue with its services. Unfortunately Vodafone Fiji's systems were not able to deliver the information for the task.

The company lacked the computing power, storage, and data management tools to analyze data quickly to make informed decisions about which deals to offer to which customers and the right times to do so. The data were stored on premises in multiple databases. Making matters worse, Amalgamated Telecom Holdings (Vodafone Fiji's major shareholder) had recently acquired several other telecommunications companies serving nearby markets in Samoa, American Samoa, Vanuatu, Cook Islands, and Kiribati. The data to be managed and mined for insights increased threefold. All those businesses had different types of data in different systems and different formats.

Vodafone Fiji's chief commercial officer Ronald Prasad and his team calculated that it would cost the company about U.S. \$2.5 million to upgrade the company's on-premises systems. They chose Oracle Autonomous Data Warehouse and Oracle Analytics Cloud services as a much more suitable and cost-effective solution. Oracle Autonomous Data Warehouse Cloud (ADWC) is a fully managed, high-performance elastic cloud service providing analytical capability over data stored in an Oracle database. The environment is optimized for data warehouse workloads and supports all standard SQL and business intelligence (BI) tools. The Oracle Autonomous

Data Warehouse provides an easy-to-use, fully autonomous database that scales elastically as workloads increase, delivers fast query performance, and requires no database administration. Vodafone Fiji would be able to extract, move, and transform data from disparate sources in the cloud, where it could be analyzed much more rapidly.

By automating many of the routine tasks required to manage Oracle databases, Oracle Autonomous Database can free up database administrators (DBAs) to do higher-level and more strategic work. The warehouse system handles a great deal of tedious technical work on its own. Vodafone Fiji would not need to hire people with specialized database management skills, which would be challenging in a small job market such as Fiji's. An on-premises data warehouse would have taken the company two months to implement, whereas it set up Oracle Autonomous Data Warehouse Cloud within 30 minutes and reporting functionality within one week. Vodafone Fiji pays only for computing resources consumed.

Oracle Analytics Cloud is a cloud-based platform that can take data in almost any form or size from almost any source (desktop, enterprise, data center), with capabilities for collecting, consolidating, and transforming data, and creating transactional and analytical reports and dashboards. Oracle Analytics Cloud provides self-service capabilities for users to perform what-if modeling and analysis. Users are empowered to visualize and discover data, including working with Big Data.

Vodafone Fiji can now easily obtain insights from the data collected and adjust its promotions to changing market dynamics. It is able to target customers who are at risk of cancelling Vodafone services as well as those most likely to respond to a special offer. The warehouse system can produce reports on call patterns to make sure customers are signed up for the optimal set of services, or target customers with 4G plans with free trials to entice them into subscribing to a data plan. Promotional campaigns customized to specific cell sites can encourage customers to get online during low utilization periods.

More than 50 rich reports are available to decision makers within minutes. Queries can be completed 5 times faster than before. A data mining procedure that used to take 125 minutes takes only 25 minutes

with Oracle Autonomous Data Warehouse; an aggregation query that used to take 294 seconds with the old on-premises system takes only 5 seconds with Oracle Autonomous Data Warehouse. Examining customer service call data, warehouse analytic tools have been able to identify trivial, repetitive inquiries where automated responses are possible, reducing inbound calls by 8 to 10 percent and freeing up

customer service agents to focus on up-selling and cross-selling based on specific customer profiles. Prasad's same three-member team has been able to handle quadruple the preacquisition workload very effectively.

Sources: www.vodafone.com.fj, accessed January 12, 2020; Tara Swords, "Call to Action," *Profit Magazine*, Fall 2019; and www.oracle.com, accessed January 12, 2020.

CASE STUDY QUESTIONS

1. Define the problem faced by Vodafone Fiji. What management, organization, and technology factors contributed to the problem?
2. Evaluate Oracle Autonomous Data Warehouse and Oracle Analytics Cloud as a solution for Vodafone Fiji.
3. How did the new Oracle tools change decision making at Vodafone Fiji?
4. Was using cloud services advantageous for Vodafone Fiji? Explain your answer.